

LOGIC AND METHODS OF PROOF

1.1 Propositions

A proposition (statement) is a sentence that is either true or false (but not both).

Examples of propositions are:

(1) John was born on 20th February 2019. (2) $3+6 = 11$. (3) $\sqrt{2}$ is irrational.

Examples of non-propositions are:

(1) What is the date today? (2) How old are you? (3) This statement is true.

There are two types of statements/propositions:

1. Atomic/simple propositions: These cannot be divided into smaller propositions.
We use capital letters to denote simple propositions.
Examples include: (1) John's leg is broken (2) 5 is a prime number (3) $\sqrt{2}$ is irrational
2. Compound propositions: These can be broken down into smaller propositions. They are constructed by using connectives.
Examples include: (1) $3 \leq 7$ (2) n^2 is odd whenever n is an odd integer

1.2 Connectives

Connectives are symbols used to construct compound statements/propositions from simple statements.

The most commonly used connectives are:

	Connective	English Meaning	Symbol
1	Conjunction	and/but/yet, Although	$\wedge$
2	Disjunction	or	$\vee$
3	Implication	If ... then	$\Rightarrow$
4	Biconditional	If and only iff	$\Leftrightarrow$
5	Negation	Not	$\neg$

Examples

1) PAQ means P and Q. 2) $P \Rightarrow Q$ means if P then Q. 3) $P \vee Q$ means P or Q

Example 1.1

Write the following compound statements into symbolic form

1) Jim is a lawyer but he is not a crook

Solution

Let P = Jim is a Lawyer and Q = Jim is a crook

1.3 Truth tables and truth values

A truth table is a convenient device to specify all possible truth values of a given atomic or compound propositions. We use truth tables to determine the truth or falsity of a compound statement based on the truth or falsity of its constituent atomic propositions.

We use T or 1 to indicate that a statement is true and F or 0 to indicate that the statement is false. So 1 and 0 are called truth values.

Truth tables for connectives

1. Conjunction

Let P and Q be two propositions, the proposition $P \wedge Q$ is called the conjunction of P and Q. $P \wedge Q$ is true if and only if both P and Q are true.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

2. Disjunction

Let P and Q be two propositions, the proposition $P \vee Q$ is called the disjunction of P and Q. The proposition $P \vee Q$ is true if and only if at least one of the atomic propositions is true. It is only false when both atomic propositions are false.

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

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3. Implication

Let P and Q be two propositions, the proposition $P \Rightarrow Q$ is called the implication of P and Q. The proposition $P \Rightarrow Q$ simply means that P implies Q. P is called the hypothesis (condition or antecedent) and Q is called the conclusion (consequence). There are many ways of stating that P implies Q.

- If P then Q
- Q if P
- P is sufficient for Q
- Q is necessary for P
- P only if Q
- Q whenever P

$P \Rightarrow Q$ is only false when P is true and Q is false.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

4. Biconditional

Let P and Q be two propositions, the proposition $P \Leftrightarrow Q$ is called the biconditional of P and Q. It simply means that $P \Rightarrow Q$ and $Q \Rightarrow P$. It is called biconditional because it represents two conditional statements. There are many ways of stating that P implies Q.

- P if and only if Q (P iff Q)
- P implies Q and Q implies P
- P is necessary and sufficient for Q
- Q is necessary and sufficient for P
- P is equivalent to Q

The proposition $P \Leftrightarrow Q$ is true when P & Q are both true and if P & Q are both false

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

5. Negation

Let P be a proposition, the proposition $\neg P$ meaning “not P” is used to denote the negation of P. If P is true then $\neg P$ is false and vice versa.

P	$\neg P$
T	F
F	T

Example 1.2

Let P and Q be propositions. Construct the truth table for the proposition $(P \wedge Q) \Rightarrow (P \vee Q)$

Solution

P	Q	$P \wedge Q$	$P \vee Q$	$(P \wedge Q) \Rightarrow (P \vee Q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

Exercise 1.2

Let P, Q and R be propositions. Construct the truth tale for the proposition $\neg(P \wedge Q) \vee R$.

1.4 Tautology, contradictions and Equivalence

Definition: A compound proposition is a tautology if it is always true irrespective of the truth values of its atomic propositions. If on the other hand a compound statement is always false regardless of its atomic propositions, we say that such proposition is a contradiction. Theorems in mathematics are always true and are examples of tautologies.

Example 1.3

The statement $P \vee \neg P$ is always true while the statement $P \wedge \neg P$ is always false

P	$\neg P$	$P \vee \neg P$	$P \wedge \neg P$
T	F	T	F
F	T	T	F

Exercise 1.3

1. Let P, Q and R be proposition. Use truth tables to prove that:

- $\neg(P \wedge Q) \equiv P \vee \neg Q$
- $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$
- $P \Rightarrow Q \equiv \neg P \vee Q$
- $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$
- $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$
- $(P \vee Q) \vee R \equiv P \vee (Q \vee R)$

Hint: To prove for example a), show that $\neg(P \wedge Q) \Rightarrow P \vee \neg Q$ is a tautology (always true).

2. Use mathematical logic to discuss the validity of the following argument:

"If girls are beautiful, they are popular with boys. Ugly girls are unpopular with boys. Intellectual girls are ugly. Therefore, beautiful girls are not intellectual."

1.5 Open sentence and quantifiers

Definition: An open sentence (also called a predicate) is a sentence that contains variable(s) and whose truth or falsity depend(s) on the value(s) assigned for the variable(s). We represent open sentences with capital letters followed by the variable(s) in the bracket. Eg $P(x)$, $Q(x, y)$

Definition: The collection of all allowed values of the variable(s) in an open sentence is called the universe of discourse.

Examples of open sentence are:

- 1) $x + 4 \leq 9$ 2) It has 4 legs 3) $x < y$

Universal quantifier-($\forall$): To say that $P(x)$ is true for all x in the universe of discourse we write $(\forall x)P(x)$. $\forall$ is called the universal quantifier.

$\forall$ means $\begin{cases} \text{for all} \\ \text{for every} \\ \text{for each} \end{cases}$

Existential quantifier ($\exists$): To say that there is (at least one) x in the universe of discourse for which $P(x)$ is true we write $(\exists x)P(x)$. $\exists$ is called the existential quantifier.

$\exists$ means $\begin{cases} \text{there is} \\ \text{there exist} \\ \text{for some} \end{cases}$

NB: Quantifying an open sentence makes it a proposition.

Exercise 1.4

1. Translate the following statements using word notation
 - a) $(\exists y)(\forall x)P(x, y)$
 - b) $(\exists x)\neg P(x)$
 - c) $(\exists x)(\exists y)P(x, y)$
 - d) $(\forall x)(\exists y)P(x, y)$
2. Write the following statements using qualifiers
 - a) For each real number $x > 0$, $x^2 + x - 6 = 0$
 - b) There is a real number $x > 0$ such that $x^2 + x - 6 = 0$
 - c) The square of any real number is non-negative
 - d) For each integer x , there is an integer y such that $x + y = -1$
 - e) There is an integer x such that for each integer y , $x + y = -1$

1.6 Negation of a quantifier

To negate a statement that involves the quantifiers $\forall$ and $\exists$, change $\forall$ to $\exists$ and $\exists$ to $\forall$ then negate the open sentence.

Examples

- 1) $\neg[(\forall x)P(x)] \equiv (\exists x)\neg P(x)$
- 2) $\neg[\text{all birds can fly}] \equiv [\text{there is at least one bird that does not fly}]$
- 3) $\neg[(\forall x)(\exists y)P(x, y)] \equiv (\exists y)(\forall x)\neg P(x, y)$

1.7 Overgeneralization and Counterexamples

Overgeneralization occurs when a pattern searcher discovers a pattern among finitely many cases and then claims that the pattern is general (when in fact it is not). To disprove a general statement such as $(\forall x)P(x)$, we must exhibit one x for which $P(x)$ is false. i.e. $(\exists x)\neg P(x)$.

Examples

- 1) Statement $(\forall x \in \mathbb{R})(x < x^2)$ is false, $x = \frac{1}{2}$ is a counterexample since $\frac{1}{2} \nless (\frac{1}{2})^2$.
- 2) For all real numbers x and y $|x+y| = |x|+|y|$. This statement is false, the counterexample is when $x=1$ and $y=-1$ since $|1+(-1)| \neq |1|+|-1|$ or $0 \neq 2$.
- 3) For all prime numbers p , $2p+1$ is prime. This statement is false, the counterexample is $p=7$.

1.8 Methods of proof in Mathematics

A proof is a process of establishing the truth of an assertion, it is a sequence of logical sound arguments which establishes the truth of a statement in question.

1) Direct Method

Suppose $P \Rightarrow Q$, in this method we assume that P is true and proceed through a sequence of logical steps to arrive at the conclusion that Q is also true.

Example 1.4

- a) Show that if m is an even integer and n is an odd integer then $m+n$ is an odd integer.

Assume that m is an even integer and n is an odd integer.

Then $m = 2k$ and $n = 2\ell + 1$ for some integer k and ℓ .

Therefore $m + n = 2k + 2\ell + 1 = 2(k + \ell) + 1 = 2d + 1$ for some integer $d = k + \ell$. Since d is an integer then $m+n$ is an odd integer.

- b) Show that if n is an even integer, then n^2 is also an even integer.

Assume that n is an even integer, then $n = 2k$ for some integer k .

Now $n^2 = (2k)^2 = 4k^2 = 2(2k^2) = 2c^2$ for some integer $c = 2k$. Since c is an integer the n^2 is an even integer.

2) Contrapositive Method

Associated with the implication $P \Rightarrow Q$ is the logical equivalent statement $\neg Q \Rightarrow \neg P$, the contrapositive of the conditional statement $P \Rightarrow Q$. So, one way to prove the conditional statement $P \Rightarrow Q$ is to give a direct proof of the contrapositive statement $\neg Q \Rightarrow \neg P$. The first step is to write down the negation of the conclusion then follow it by a series of logical steps that this leads to the negation of the hypothesis of the original conditional statement.

Example 1.5

- a) Show that if n^2 is even integer then n is an even integer.

We will show that if n is odd then n^2 is odd. Assume that n is odd. Then $n = 2k + 1$ for some integer k .

Now $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2d + 1$ for some integer $d = 2k^2 + 2k$. Since d is an integer then n^2 is an odd integer. Thus, by contrapositive if n^2 is even then n is itself even.

b) Show that if $2n$ is odd integer then n is an odd integer.

By contrapositive, we show that if n is even integer then $3n$ is an even integer.

Now $n = 2p$ and $3n = 3 \cdot 2p = 2(3p) = 2c$ for some integer $c = 3p$. Since c is an integer then $3n$ is even and so by contrapositive, if $3n$ is odd then n is odd

3) Contradiction Method

Assume that P is true and Q is false i.e. $P \wedge \neg Q$ is true. Then show a series of logical steps that leads to a contradiction or impossibility or absurdity. This will mean that the statement $P \wedge \neg Q$ must have been fallacious and therefore, its negation $\neg[P \wedge \neg Q]$ must be true. Since $\neg[P \Rightarrow Q] \equiv [P \wedge \neg Q]$, it follows that $[P \Rightarrow Q] \equiv \neg[P \wedge \neg Q]$ and hence $P \Rightarrow Q$ must be true.

Definition: A real number r is said to be rational if there are integers n and m ($m \neq 0$) such that $r = \frac{n}{m}$ with greatest common divisor between $[n, m] = 1$. We denote the set of rationals by $\mathbb{Q}$. A real number that is not rational is said to be irrational.

Example 1.6 (proofs by contradiction)

a) Show that $\sqrt{2}$ is irrational. That is there is no integer p and q such that $\sqrt{2} = \frac{p}{q}$ or there is no rational number p such that $p^2 = 2$.

By contradiction, we assume that $\sqrt{2}$ is rational.

So, we have $\sqrt{2} = \frac{p}{q}$. And so $2 = \frac{p^2}{q^2}$ thus $2q^2 = p^2$ (*)

From (*), observe that p^2 is even and so p is even and can be written as $p = 2c$ for some integer c .

So (*) becomes: $2q^2 = (2c)^2 = 4c^2$ and so $q^2 = 2c^2$ (**)

Similarly, from (**) it is observed that q^2 is even and so q is itself even. Therefore, both p and q are even integers; implying that they both have a common factor of 2. This contradicts the fact that the GCD $[p, q] = 1$ and thus $\sqrt{2}$ is not rational but rather it is irrational.

b) Show that if $3n$ is an odd integer then n is an odd integer.

We will use contradiction. Assume that $3n$ is an odd integer and n is even integer.

Then $3n = 2k+1$ and $n = 2d$ for some integers k and d .

Therefore $3n = 2k+1 = 2(3d) = 2c$ for some integer c . Since c is an integer then $3n$ is an even integer which indicates that $3n$ is both odd and even which is impossible. Thus, if $3n$ is odd n cannot be even but rather n is odd